WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER PROGRAM

AZNS-122 (PAD #55), TYPE: IIF

Drilling Department

JANUARY 2024

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

This Document covers the work over of well AZNS-122. Main headlines of program are de-complete the well and run new completion string.



Figure 1: AZNS Oilfield Location Map

1.2 Well Information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Type Of Operation:	Workover
Well Name / Type :	AZNS-122 / Fahliyan well in Block II
Pad Number:	55
Well Type:	Vertical
Wellhead Coordinate:	N: 3,447,673 E: 194,074
Fahliyan E.P. Coordinate:	N: 3,447,664.8 E: 194,078.6
TD Coordinate:	N: 3,447,666.83 E: 194,075.98
Well Objective:	De-complete the well, Re-complete the well, Due to abnormal annulus pressure in 9-5/8" CSG X 4-1/2" TBN annulus, find leakage point & cure it
RTE:	13.45 m
GLE:	3.95 m
Well Duration :	45 (Main Plan) + 7 (Contingency Plan)
Total Depth (MD):	4208 m
Producer Formation:	Perforated in 7" liner (Exiting Perforation 4086 – 4120m)
Formation TD :	Fahliyan
Reservoir :	Fahliyan

Table 1: AZNS-122 information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.
Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 2: Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.
Block II is located between well AZN-010 and AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal
Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

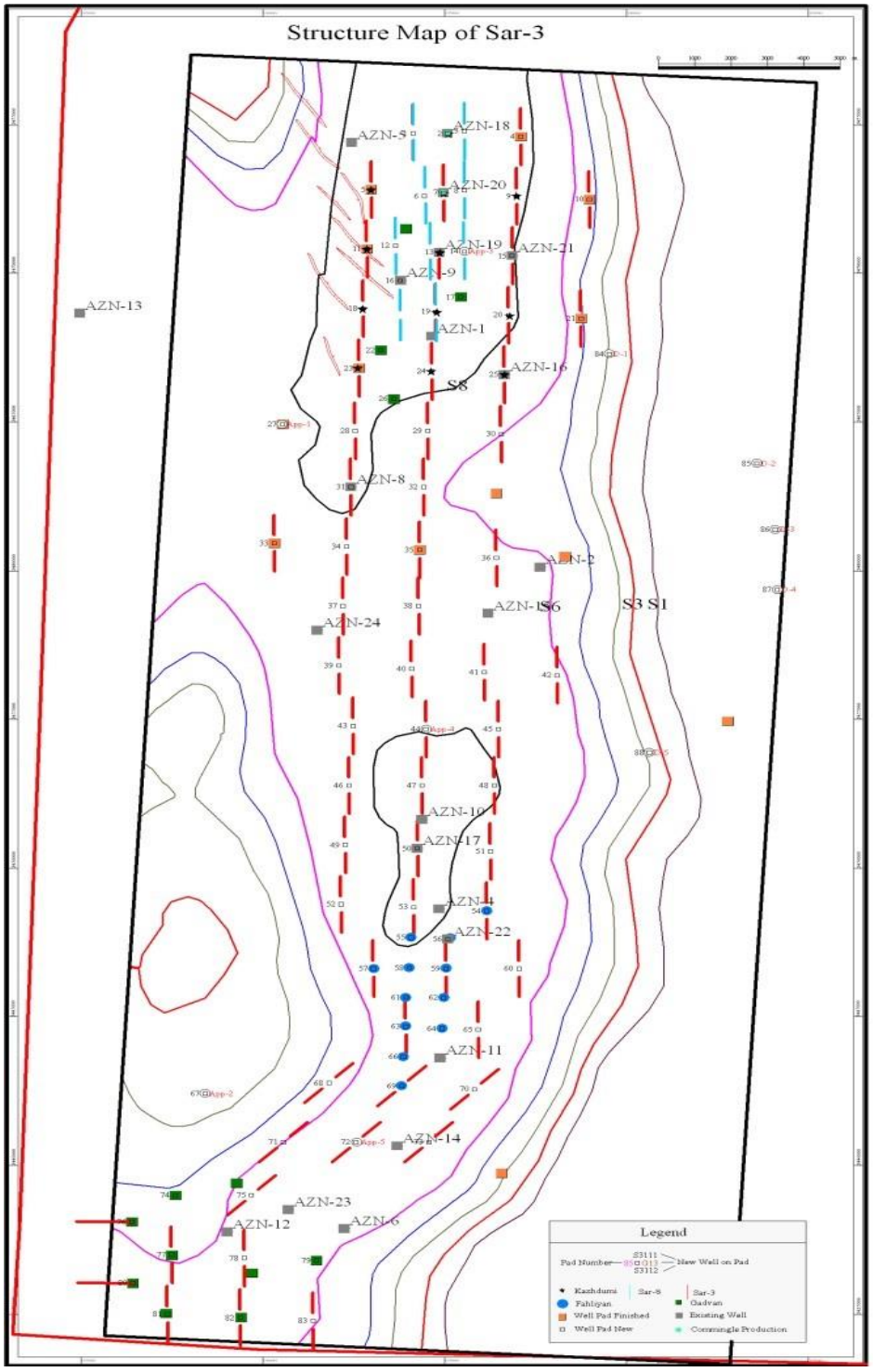


Figure 2: AZNS Area Distribution

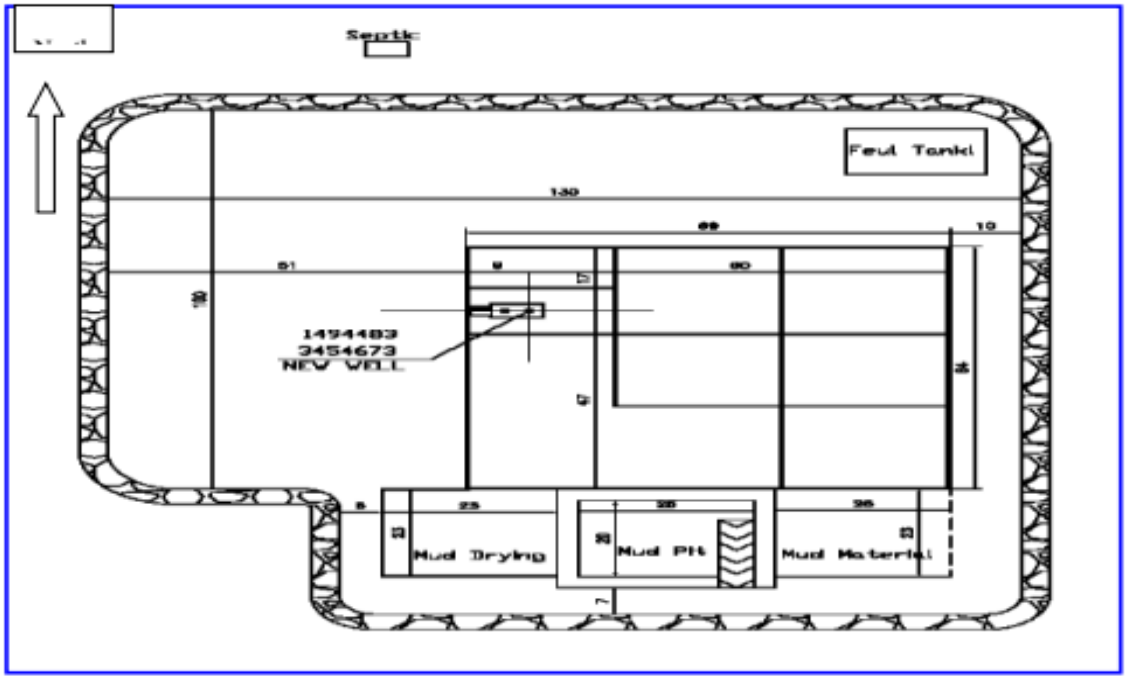
1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

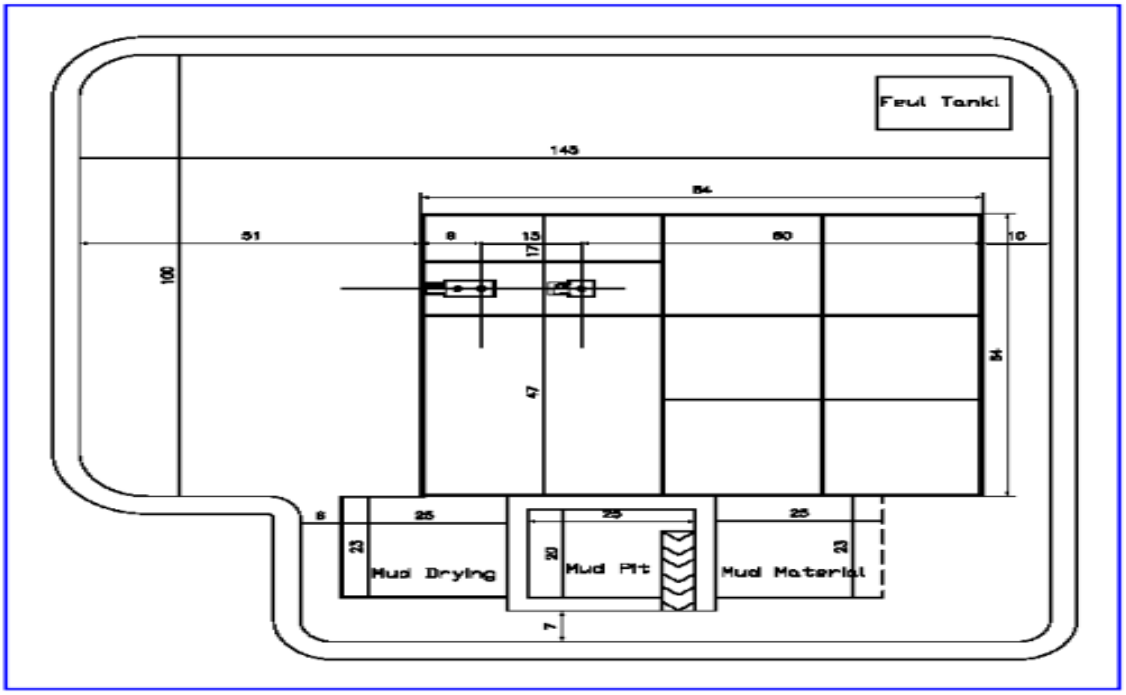
Table 3: Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm2
• Fuel tank concrete pond dimension: Length 20m×Width 12m×Height 0.5m
• Drainage: Width 30cm × Depth 30cm
• Burning pit: 15m×15m square, located on South-East or South-West with at least 350m distance from well head
• Cellar to cellar distance in multi-well well pad: 15m
• Conductor pipe depth under cellar bottom is 5.5m. The conductor pipe will be driven while constructing the well pad.
• Mud material dimension is 25m*35m +granular surface
• Mud drying dimension is 25*35m + concrete surface
• Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

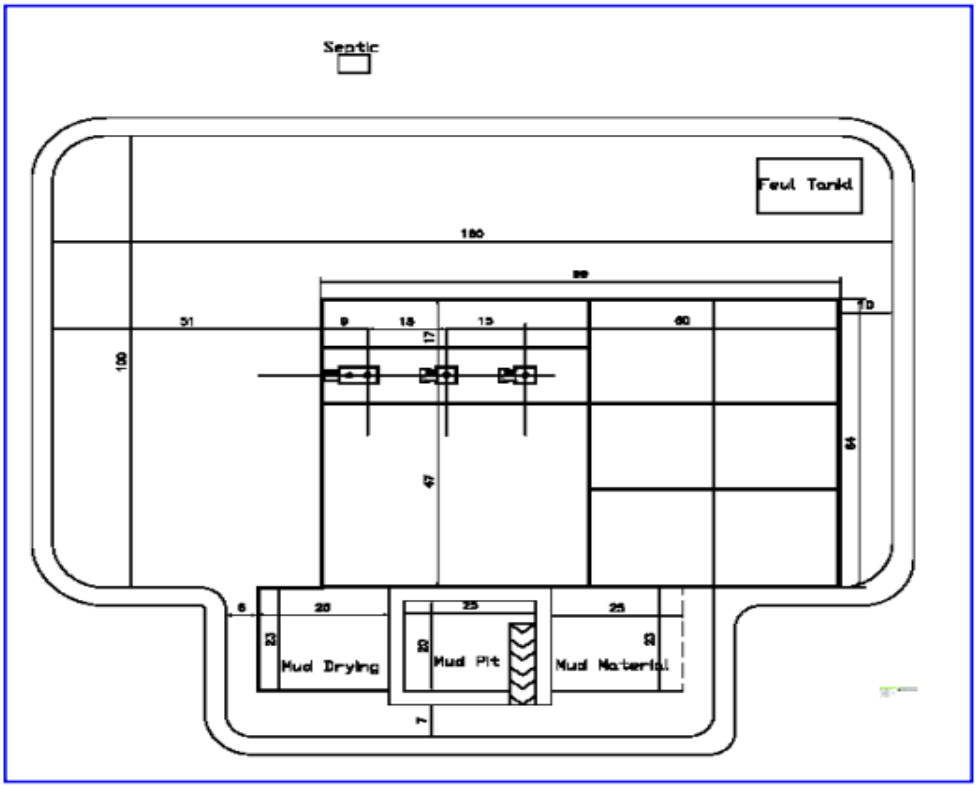
1. Types of well pads:
• One well in one pad.
• Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
• Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.



Well Pad for one well



Double-well Well Pad



Triple-well Well Pad

Figure 3: Well Pads Sketch

1.5 Reservoir Data

Reservoir Name:	Fahliyan	Unit
Fluids:	Oil	….
Datum Depth:	-4000	M,TVDss
Datum Pressure:	Fah-4a= max 6800 psi Fah-5a= max 7400 psi	Psi
Pressure Gradient:	0.28	Psi/ft
Temperature:	132	C°
Temperature Gradient:	N/A	1.32 F/100ft 0.73 C/100ft 2.4 C/100m
H2S:	H2S in Gas at Atm condition 3000 PPM	PPM
CO2:	Co2 in Gas at Atm Condition 50000 PPM	PPM
Salt Salinity :	143000	ppm
Calcium Content:	8750	(mg/lit)

Table 4: Reservoir Data

Reservoir	Sub-layer	Mid of Sub-LayerDepth in AZNS-122 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure
Fahliyan	Fah-1	-3993	257	1.32	8946	Dynamic Model
	Fah-2a	-4034	259		8780	Dynamic Model
	Fah-2b	-4048	260		8795	Pressure Gradient
	Fah-3a	-4058	260		8036	Dynamic Model
	Fah-3b	-4066.5	261		8794	Pressure Gradient
	Fah-4a	-4076	261		6380	Dynamic Model
	Fah-4b	-4093	262		6944	Dynamic Model
	Fah-5a	-4109	262		7480	Dynamic Model
	Fah-5b	-4140	264		7500	Dynamic Model

Table 5: Predicted Pressure and Temperature and TTZ Contact

1.6 Formation Pressure Prediction in Block II

Predicted Formation Pressure in Block II				
Formation	**Interval**	**Pore Pressure**	**Collapse Pressure (Break-Out Pressure)**	**Fracture Pressure (Break-Down Pressure)**
	m	*SG*	*SG*	*SG*
Aghajari	0-1095.9	0.98-1.18	0.06-1.39	1.01-1.89
Gachsaran	1095.9-1461	1.13-1.31	0.78-1.36	1.89-2.01
Asmari	1461-1728	1.14-1.26	0.6-1.28	1.43-1.79
Pabdeh	1728-2196	1.20-1.26	0.82-1.36	1.40-1.78
Gurpi	2196-2536	1.21-1.26	0.66-1.02	1.40-1.76
Ilam	2536-2626	1.18-1.24	0.20-0.9	1.41-1.72
Laffan	2626-2633	1.13-1.18	0.24-1.11	1.41-1.72
Sarvak	2633-3278	1.14-1.26	0.24-1.23	1.44-1.81
Kazhdumi	3278-3481	1.1-1.14	0.46-0.96	1.93-1.94
Dariyan	3481-3647	1.18-1.25	0.55-1.1	1.96-2.06
Gadvan	3647-3937	1.1-1.25	0.55-1.44	1.97-2.11
Fahliyan	3937-T.D.	1.60-1.3	1.20-1.75	2.08-2.12

Table 6: Formation Pressure Prediction

1.7 Drilling History

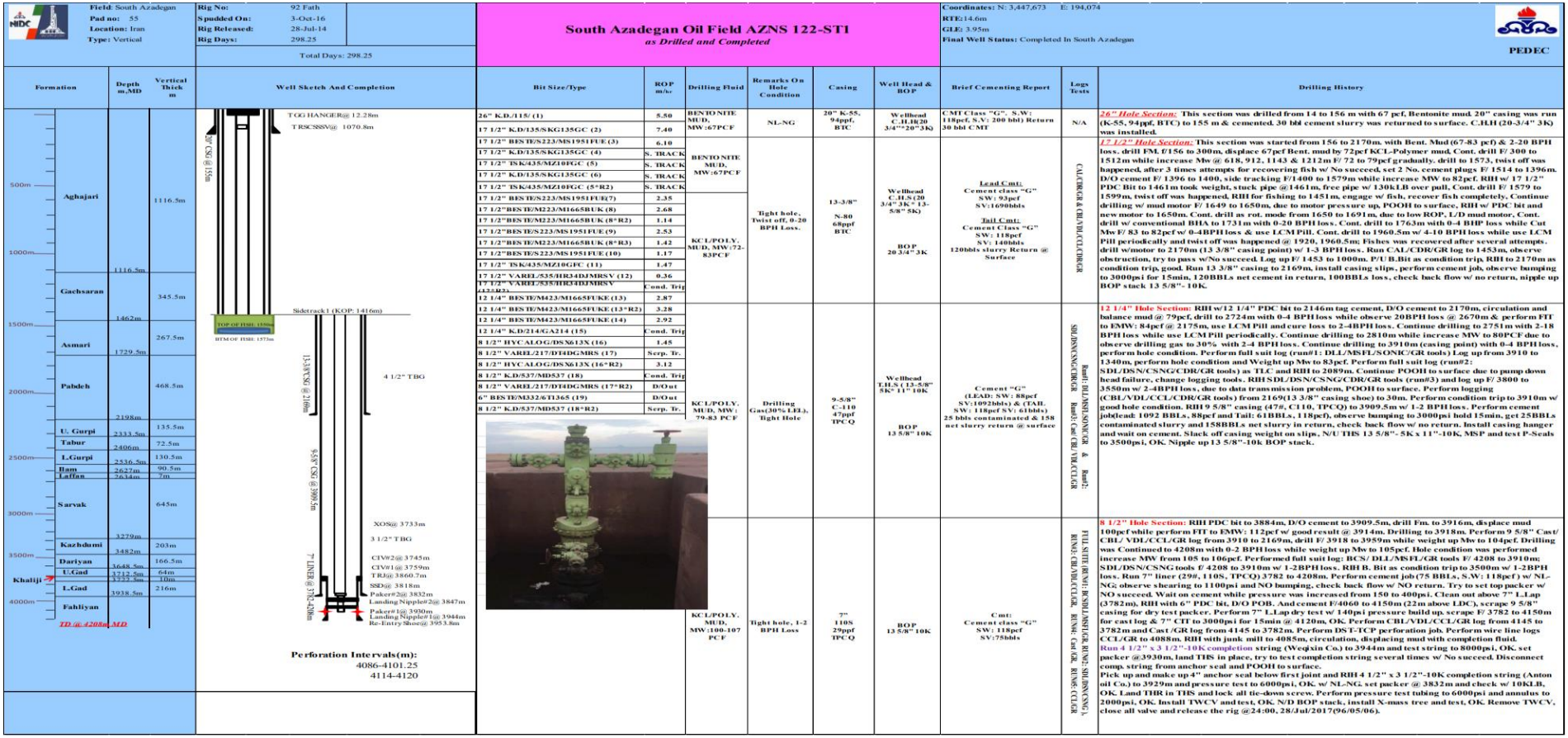
The AZNS 122-ST1/F02 (Pad 55) was drilled by rig 92 Fath, which rig moving was spudded on 3-October-2016 @ 18:00.

- 26" hole was spudded @ 21:00 on 3-October-2016 & drilled to casing point @ 156m, 09:30 on 5-October-2016, Mw 67pcf.
- 20" casing was run to 155m & cemented by inner stab-in string with slurry to surface.
- 17-1/2" hole was started @ 02:30 on 10-October-2016 & drilled to 2170m @ 21:00 on 11-Dec-2016, Mw 67 – 83pcf.
- In 17-1/2" hole section, drill string was twist off @ 1573m, tried to recovered fish with no succeed, set cement plug & side track the well, KOP @ 1416m.
- In 17-1/2" hole section, drill string was twist off @ 1599, 1920 & 1960.5, fishing operation was performed & fish was recovered in each operation.
- 13-3/8" casing was run & cemented @ 2169m with slurry to surface.
- The 12-1/4" hole section was started @ 08:00 on 20-Dec-2016 & drilled to 3910 @ 07:00 on 20-Jan-2017, Mw 79 – 83pcf.
- 9-5/8" casing was run & cemented @ 3909.5m with slurry to surface.
- The 8-1/2" hole section was started @ 17:00 on 08-Feb-2017 & drilled to 4208m @ 24:00 on 16-Feb-2017, Mw 100 – 107pcf.

1. 7" liner was run & cemented f/3782.2 to 4208m, the dry test result shows that the liner lap is OK.

- 2192.5 hrs. Waiting for receive completion fluid.
- Displace mud in hole with completion fluid.
- 4-1/2"*3-1/2"-10K Completion string (Weqixin Co.) was run, packer set @ 3930m, try to test completion string several times with no succeed, disconnect completion string from anchor seal & POOH to surface. P/U & M/U 4" anchor seal below first joint & RIH 4-1/2"*3-1/2"-10K completion string (Anton Oil Co.), set packer @ 3832m, perform pressure test was OK, install X-mass tree & test, OK.
- The rig 92 Fath was released @ 24:00 on 08-July-2017. (1396/05/06)

1.8 Well Montage



1.9 Casing & Liner Depth

Casing Spec.	20"	K-55, 94#, BTC	0-155 m
	13-3/8"	N-80, 68#, BTC	0-2169 m
	9-5/8"	C-110, 47#, TPCQ	0-3909.5 m
	7"	110-S, 29#, TPCQ	3782 – 4208 m

Table 7: Casing & Liner Data

1.10 Well Schematic before Workover

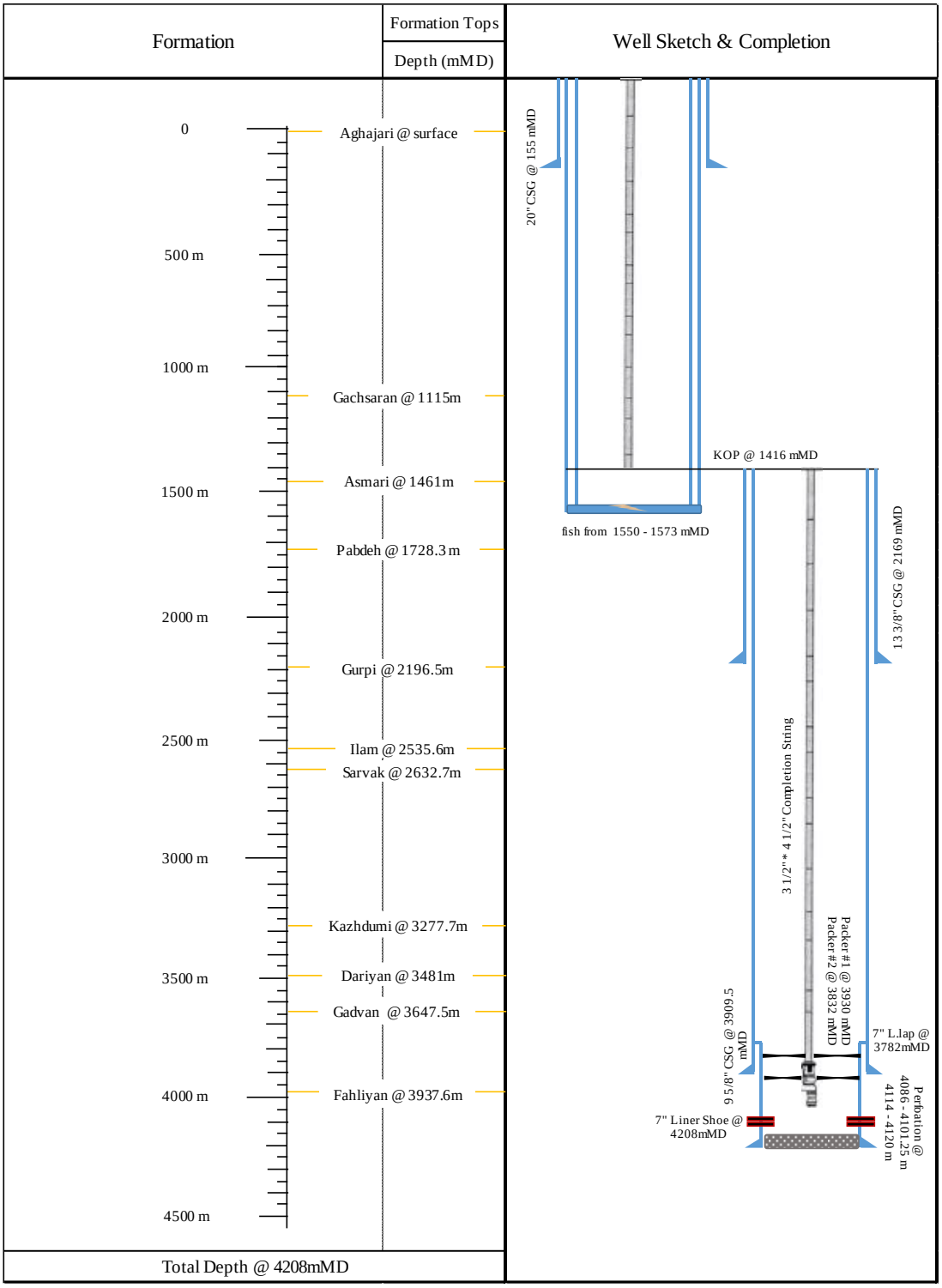


Figure 4: Well Schematic Before workover

1.11 Well Schematic after Workover

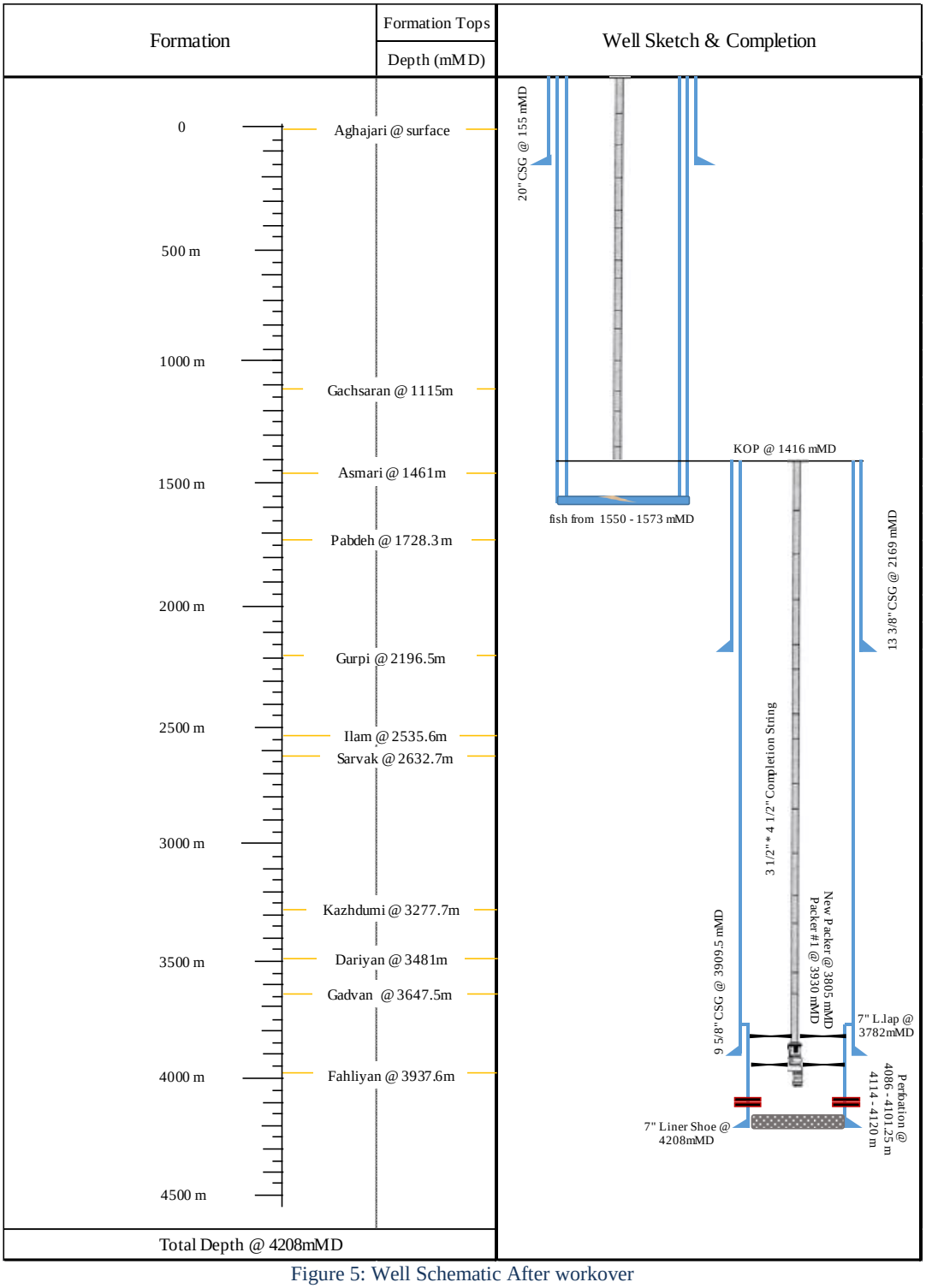


Figure 5: Well Schematic After workover

2. Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
		Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
Miocene							
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi Tarbur L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
		Laffan		Claystone	2691	2642	2625
Turonian Cenomanian		Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian		Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian		Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian		Fahliyan		Limestone, Marl	4092	3989	3974

Figure 6: Azadegan Stratigraphic Column

3. Logging Program

Hole Size	Casing Size	Formation	Logs
			Cased Hole
12-1/4"	9-5/8" CSG.	Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan	Corrosion Log
8-1/2"	7"L.	Fahliyan	Corrosion Log

Table 8: Logging Program

4. Program Outline

4.1 Contractors Involved

Table 9: Contractor Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 208
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	MAPSA KISH
6	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC

4.2 Time Breakdown

De-Complete, Re-Complete Operation TBD of Well: AZNS-122 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over									
Held PJSM. Spud the Rig. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC.	Trip	4208	30	1.25	1.3	1.4	33.0	1.38	1.38
Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status.	Wellhead	4208	15	0.63	1.9	0.7	16.5	0.69	2.1
Secure the well by close 3SV, Remove 7-1/16" XMT cap, R/U Slick line,RIH w/Gauge Cutter to top of 3SV, RIH & set protection sleeve, RIH w/Gauge Cutter to top of SSD if not ok follow Note: Chemichal Wash May Be Needed While RIH W/CT.	Trip	4208	30	1.25	3.1	1.4	33.0	1.38	3.4
R/U S/L. RIH with Shifting Tools opesn SSD.POOH S/L.displace the well with heavy mud.Retrieve Protection Sleeve, Close 3SV, R/D S/L.	Trip	4208	48	2.00	5.1	2.2	52.8	2.20	5.6
Set BPV inside tubing hanger & pressure test, N/D XMT & N/U BOPs & pressure test same.	Wellhead	4208	50	2.08	7.2	2.3	55.0	2.29	7.9
Retrieve BPV, R/U casing running equipment, RIH with TBH retrieving tool & engage with TBH, apply over pull to un-latch the seal assembly. Note: If unable to un-latch seal assembly, follow contingency plan #1.	Completion	4208	48	2.00	9.2	2.2	52.8	2.20	10.1
Perform BOP Test.	BOP	4208	8	0.33	9.5	0.4	8.8	0.37	10.5
Start circulation immediatley after stinging out seal assy & observe well.	Completion	4208	16	0.67	10.2	0.7	17.6	0.73	11.2
POOH & L/D completion string.	Trip	4208	96	4.00	14.2	4.4	105.6	4.40	15.6
RIH with mill assembly & mill Upper packer, chase to bottom, POOH.	Trip	4208	72	3.00	17.2	3.3	79.2	3.30	18.9
Perform BOP Test.	BOP	4208	8	0.33	17.5	0.4	8.8	0.37	19.3
RIH with taper tap & catch the packer, recover the completion packer, POOH, L/D BHA.	Trip	4208	72	3.00	20.5	3.3	79.2	3.30	22.6
RIH with 9-5/8" * 7" tandem scraper BHA, Scrap RTTS Packer setting depth, CBU & POOH to surface.	Trip	3832	48	2.00	22.5	2.2	52.8	2.20	24.8
RIH with wireline corrosion logging tools to bottom, log up to surface in two separate run (7" Liner & 9-5/8" Casing).	Log	4208	48	2.00	24.5	2.2	52.8	2.20	27.0
P/U & M/U RIH w/ 7" RTTS Packer and Perform 7" Liner Lap Positive Test. Note: If Positive test is not ok, follow contingency plan #2.	Dry Test	4208	20	0.83	25.4	0.9	22.0	0.92	27.9
RIH with 5-7/8" mill & clean out BHA (7" scraper included) to top of lower packer @3930 m, displace the well with completion fluid (non-solid), POOH & L/D DP's.	Trip	3832	68	2.83	28.2	3.1	74.8	3.12	31.0
Run 10K completion string as per program. NOTE: Annulus pressure test during completion operation is related to corrosion log result.	Completion	4208	154	6.42	34.6	7.1	169.4	7.06	38.1
Set BPV inside TBH, N/D BOP's & install 10K XMT.	Completion	4208	60	2.50	37.1	2.8	66.0	2.75	40.8
Pressure test the XMT.	Completion	4208	16	0.67	37.8	0.7	17.6	0.73	41.6
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.	Completion	4208	46	1.92	39.7	2.1	50.6	2.11	43.7
Retrieve BPV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
RIH with CT to bottom, displace the well with diesel & observ well, POOH.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
RIH with C/T & perform acid wash as per program, POOH to surface.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7

De-Complete, Re-Complete Operation TBD of Well: AZNS-122 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over									
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
RIH with S/L, retrieve protection sleeve.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion	4208	Rig Less	0.00	39.7	0.0	0.0	0.00	43.7
Total Duration of Work-over:				***39.7***	***39.7***			***43.7***	***43.7***
NOTE:									
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip	4208	6	0.25	40.0	0.3	6.6	0.28	44.0
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service	4392	22	0.92	40.9	1.0	24.2	1.01	45.0
Total Duration of AZNS-122 Work-over Well:		**4208**			**40.9**				**45.0**
Contingency Plans TBD of Well: AZNS-122 Work-over	**IADC Code**	**Depth (meters)**	**Time Duration (Hours)**	**Time Duration (Days)**	**Total Duration (Days)**	**Contingency (%) 10**	**Time Duration w/ contingency (Hours)**	**Time Duration w/ contingency (Days)**	**Total Duration w/ contingency (Days)**
Based on operation situation in each step, the following contingency plans might be followed									
Contingency plan #1: RIH w/ colliding tools & collide tubing f/ top of packer, then mill tubing & packer.	Completion	4208	117.0	4.9	4.9	5.4	128.7	5.4	5.4
Contingency plan #2: RIH & install liner top packer on top of 7" liner lap.	Trip	4208	36.0	1.5	6.4	1.7	39.6	1.7	7.0
Total Duration of AZNS-122 Work-over Well:		**4208**			**6.4**				**7.0**

5. Sequence of Operation

5.1 Main Objective

Decision has been made to de-complete he well & re-complete with new completion equipment.

5.2 Main plan

1. Held PJSM. Spud the Rig. Stand back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC.
2. Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status.
3. Secure the well by close 3SV, Remove 7-1/16" XMT cap, R/U Slick line, RIH w/Gauge Cutter to top of 3SV, RIH & set protection sleeve, RIH w/Gauge Cutter to top of SSD if not ok follow.

Note: Chemical Wash May Be Needed While RIH W/CT.

4. R/U S/L. RIH with Shifting Tools open SSD.POOH S/L. Displace the well with heavy mud. Retrieve Protection Sleeve, Close 3SV, R/D S/L. Set BPV inside tubing hanger & pressure test, N/D XMT & N/U BOPs & pressure test same.
5. Retrieve BPV, R/U casing running equipment, RIH with TBH retrieving tool & engage with TBH, apply over pull to un-latch the seal assembly.

Note: If unable to un-latch seal assembly, follow contingency plan #1.

6. Start circulation immediately after stinging out seal assembly & observe well.
7. POOH & L/D completion string.
8. RIH with mill assembly & mill Upper packer, chase to bottom, POOH.
9. RIH with taper tap & catch the packer, recover the completion packer, POOH, L/D BHA.
10. RIH with 9-5/8" * 7" tandem scraper BHA, Scrap RTTS Packer setting depth, CBU & POOH to surface.
11. RIH with wireline corrosion logging tools to bottom, log up to surface in two separate run (7" Liner & 9-5/8" Casing).
12. P/U & M/U RIH w/ 7" RTTS Packer and Perform 7" Liner Lap Positive Test.

Note: If positive test is not ok, follow contingency plan #2.

13. RIH with 5-7/8" mill & clean out BHA (7" scraper included) to top of lower packer @3930 m, displace the well with completion fluid (non-solid), POOH & L/D DP's.
14. Run 10K completion string as per program.

Note: Annulus pressure test during completion operation is related to corrosion log result.

15. Set BPV inside TBH, N/D BOP's & install 10K XMT.
16. Pressure test the XMT.

17. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.

Rig less Sequence of Operation

18. Retrieve BPV, R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.

19. RIH with CT to bottom, displace the well with diesel & observe well, POOH.

20. R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.

21. Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.

22. R/U SL, RIH & set protection sleeve, POOH.

23. RIH w/Gauge Cutter to top of NO GO landing nipple POOH, R/D S/L.

24. RIH with C/T & perform acid wash as per program, POOH to surface.

25. R/U SL, RIH & retrieve protection sleeve, POOH, R/D S/L.

26. Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs.

Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded).

Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.

27. RIH with S/L, set protection Sleeve, POOH & R/D SL.

28. RIH with S/L, install plug & prong into NO GO landing nipple, POOH.

29. RIH with C/T to bottom & displace the well with diesel, POOH.

30. RIH with S/L, retrieve protection sleeve.

31. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.

Contingency Plans of Well AZNS-122 Work-over:

Based on operation situation in each step, the following contingency plans might be followed:

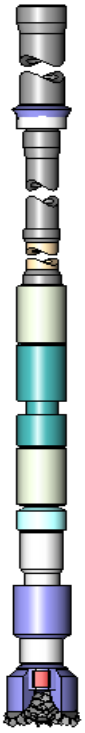
Contingency plan #1: RIH w/ colliding tools & collide tubing f/ top of packer, then mill tubing & packer.

Contingency plan #2: RIH & install liner top packer on top of 7" liner lap.

6. BHA Program

6.1 5-7/8" BHA

			Collar	OD
MILL SIZE (inch)	5-7/8"	BHA Configuration to C/Out Inside 7" Liner		
Depth In/Out (m)	4208		DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0.784			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	23,229			
Bouyed Weight Below Jar (lb)	16,462			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	5-7/8" Mill	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

7. Mud & Cement Program

Note: Detailed mud & cement program, will be issued in due time as per client approved template.

7.1 Workover Hole Section

Workover section	7" Liner
From: 3782m	TO: 4208m
Kill the well, POOH of Completion String, Complete the hole with Completion 10K & Completion Fluid	

7.1.1 Overview

Drilling Fluid System	Kill Mud and Completion Fluid
Key products	Workover Fluid , Completion Fluid, Kill Mud
Water type	Drill Water
Solid Control	Shale Shaker
Potential problems	Down Hole losses , gas/oil influx, H2S,

Interval Mud Properties						
Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
0-4208	106	30-55	8-22	NC	9-10.5	45-55
Interval Completion Fluid Properties						

Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
Surface~3780	106	ALAP	N/A	NC	6-7.5	27-30

7.1.2 Mud Discussion & Mixing Procedure

7.1.2.1 Mud type

Kill mud 106 pcf will be used to kill the well and well will be displaced by Completion fluid of CaBr2 106pcf. Attention must be paid that there is Calcium Bromide Brine inside the well and all of them must be retrieved in separated mud tank by the time of displacing the well by kill mud. (As what the hole dictates, mud properties may be varied by the time).

7.1.2.2 Drilling Fluid Formulation

Kill Mud/Work over Mud formulation		
Component	**Conc. (ppb)**	**Component Description**
Drill water	0.5766 bbl/bbl	Base Fluid
Caustic Soda	0.3	PH Modifier (lime can be used instead)
Soda Ash	0.3	Hardness Remover and
Sodium Chloride	63	Non-Solid weighting Agent
Xanthan-Gum	0.2	Viscosifier (In case of need and based on situation can be used)
Calcium Carbonate	325	Weight Agent

*Above calculation is based on mud weight 106 pcf

*Lime can be substituted for caustic soda

Completion Fluid Formulation		
Component	**Conc. (ppb)**	**Component Description**
Calcium Bromide (14.2 ppg)	0.984bbl/bbl	Base Fluid
Corrosion Inhibitor	1% Vol	Anti-Corrosion
H2S Scavenger Liquid	0.5% Vol	H2S Removal/ Anti Corrosion
Oxygen Scavenger CA	0.5 ppb	Oxygen Removal/Anto Corrosion

The performance requirements designed for the completion fluid system are as follows:

- Recovered permeability: ≥ 90 ٪
- Corrosion rate ＜0.076 mm/year
- Diameter of solid particle ≤ 2μm
- For fluid pumping in the well NTU<30
- For fluid returning from the well: NTU <150
- Acid Soluble/ Removable with Solvent

7.1.2.3 Drilling Fluid Formulation

Kill Mud Mixing Procedure:

1. Fill tanks with calculated amount of Drill Water 0.576 bbl/bbl.
2. Treat Drill Water with Caustic soda and Soda Ash by 0.3pbb.
3. Mix NaCl at 66 ppb.
4. Add Xanthan-Gum by 0.2pbb.
5. Add CaCO3-F (325ppb) to reach MW of ±106 pcf.
   - In case of foaming use Alcohol-Based Defoamer.

Note: Use Hi Vis pill to clean hole if necessary.
Calculation is based on per bbl of mud.

Completion Fluid Mixing Procedure:

1. Make sure that all pits and lines involved with completion fluid are thoroughly cleaned, flushed and free of any fluid and solid.
   - Make sure that at least one large pit is kept empty for transferring ready brine into after given time for settling. After transferring one pit to empty pit, the first one is to be flushed and cleaned for accommodating the brine of next ready pit, and so on to the last pit.
   - Depends on the rig facilities and structure, mud engineer on board may find it better to prepare the fluids pit-by-pit.
2. Transfer the content of CaBr2 drums into designated pits.
   - Record the number of drums poured in each tank for accurate volume in it as the base of calculation for adding other additives into; do not rely on pit gauge, sounding or mud logging sensor, but have a look on them for comparison and/or correlation if found useful.
3. Mix Corrosion Inhibitor into pits at concentration of 3.5pbb (1% by volume).
4. Mix H2S Scavenger Liquid into pits at concentration of 0.5% by Volume.
5. Do not add Oxygen Scavenger CA at this stage as it should be added directly from the top of the pit, just prior to pump the fluid into the well.
6. Shut down all agitators and mixing pumps, and leave the mixed fluid in the pits for at least 12 hours to let any solid particle inside (mostly insoluble Lime) settles down.
7. Start transferring the clear fluid of the first pit to the empty pit by means of an air-powered Wilden pump.
   - Make sure that the suction hose of the Wilden pump is not put deep close to the bottom of the pit, otherwise, settled solids are sucked in.

8. After finishing the first pit, stop Wilden pump, wash, flush and clean the pit, use it as the empty pit for accommodating the fluid of second pit using the and same above method.

Note: all CaBr2 brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of CaBr2.
Based on well readings and the operation, mud weight may be varied.

7.1.2.4 Downhole Losses

As perforation had been done in well AZN#122 and there is exposure to reservoir, risk of any probable formation losses must be taken into consideration.
Losses will be minimized/avoided by the planned use of properly sized bridging/weight material (and Polymer) in conjunction with appropriate drilling practices. If seepage losses are noted on the well no specific action will be taken but close monitoring of the loss rate is to be made. For moderate loss rates addition of CaCO3 50 micron and 150 micron is recommended to ensure that there is adequate correctly sized material to act as bridging agents.

Seepage losses Less than 10 bbl/hr.

No specific action will be taken, but close monitoring of the probable loss rate is made.

Losses from 10-30 bbl/hr.

- Load CaCO3-M (150-300 micron) in system up to 5 ppb

***If loss not cured and even tends to increase, then:**

- Pump 40-60 bbl of pill made by 7-10 ppb CaCO3 150-300µ & 7-10 ppb CaCO3 600µ

Losses from 30-50 bbl/hr.

- Load CaCO3-M (150-300 micron) in system up to 5 ppb
- Since some of the CaCO3-M will be discarded over the shakers, CaCO3-M to compensate the separated amount on the shakers.

***If loss not cured and even tends to increase, then:**

- Pump 40-60 bbl of pill made by
- 10-20 ppb CaCO3 200-300µ
- 10-20 ppb CaCO3 600 µ
- 10pbb Oyster Shell-F
- 10pbb Oyster Shell-M.

Losses from 50-100 bbl/hr.

- Load CaCO3-M (150-300 micron) in system up to 5 ppb
- Load CaCO3-C (600-100 micron) in system up to 5 ppb
- Since some of the CaCO3-M&C will be discarded over the shakers, add CaCO3-M&C to compensate the separated amount on the shakers.

***If loss not cured and even tends to increase, then:**

- Pump 40-60 bbl of the following pill:

- 10-20 ppb CaCO3 200-300 micron
- 10-20 ppb CaCO3 600 micron
- 7-10 ppb of each CaCO3 1000 micron & CaCO3 2000 micron
- 5-10 pbb Oyster Shell -F
- 5-10 pbb Oyster Shell -M

Losses more than 100 bbl/hr.

High concentrated LCM pill of various sizes of Calcium Carbonate and Oyster Shell-F&M&C should be pumped and spotted and gently squeezed over the loss zone.

Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30 minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close BOP and squeeze 10 bbl LCM pill into formation without exceeding 200 psi.
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50 bbl of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20 bbl of LCM pill into the formation without exceeding 200 psi.
6. If a squeeze pressure develops, hold the pressure for 1 hour before running the drill string back to bottom.

Notes:

- LCM pills must be thoroughly homogenised and viscous enough before being pumped.
- Pumping high-concentrated LCM pill is subject to use of open-end DP or Mule-Shoe.

- Remove filter screens from the drill pipe (if there is any) and if necessary, remove strainers from the mud pump suction lines.

7.1.2.5 Engineering Comments

Keep watching and control on well to record loss.
All CaBr2 brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of CaBr2.

7.1.2.6 Drilling Issues

There is no drilling objective in AZN-122 workover well; mud should be compatible with reservoir and work over operation.
Solid Control equipment

- Use shale shaker to remove debris and LCM particle in case of use.
- Personal Safety

7.1.2.7 Mud Type

Inhibited CaBr2 brine (Calcium Bromide Brine) will be applied as completion fluid. And simple salt saturated kill mud to kill the well and clean out inside 9 5/8", 7" Liner.

Displacement Design
The combined effect of scrapers being run in the hole, the high flow rate will help to remove/disperse general debris.
At least 150 m of pill/spacer coverage or 3 min contact time at the highest pump rates should be considered.
Saturated CaCl2 brine can be applied as spacer/Hi-Vis pill while displacing.
Pit levels should be monitor throughout the displacement operation to catch and prevent an influx entering the well contributing to an incident.
All CaBr2 brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of CaBr2.

No.	Description	Volume [bbl.]	Comment
1	Viscous pill/spacer	50	Clean pill

2	Displace to inhibited CaBr2 Brine at 106pcf (Completion Fluid)
3	Continue pumping Completion Fluid until cleanliness specification has been met.

All CaBr2 retrieved from well must be stored in separated tanks. All CaBr2 left on surface, even dead volume of tanks, must be transferred back to drums at end of well. Not even a drop of Cabr2 is not allowed to be wasted or dump in whole operation duration.
It is recommended to use desander tank as active pit of CaBr2 to minimize dead volume and return path.

7.1.2.8 Volume accounting & estimated required material

Work over /Kill Mud Fluid				AZADEGAN #122
Fluid Density (pcf)	106		Fluid System	Water base Work Over Fluid /Salt Saturated
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3909.5		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)	4208
Liner Depth (m)	4208		Liner Lap (m)	3782
Liner OD / ID (inch)	7	6.276	Liner Length (m)	426

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	350
Cased Hole Volue [Capacity of 3782m 9⅝" Casing + Capacity of 426m 7" Liner]	960
Total New Volume Required (bbl)	**1310**

Estimated Chemical Consumption and Cost based on 106pcf MW							
PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Drill Water	bbl	N/A	0.576 bbl/bbl	755		755	N/A
Caustic Soda	25 Kg	N/A	0.3	7	25	32	N/A
Soda Ash	50 Kg	N/A	0.3	4	16	20	N/A

Sodium Chloride	1 MT	N/A	63	37	18	55	N/A
Xanthan-Gum	25 Kg	N/A	0.2	5	20	25	N/A
Calcium Carbonate	1 MT	N/A	325	194	30	224	N/A

*Above calculation are based on normal hole condition

*Above Calculated is Based on **106pcf**

Completion Fluid				AZADEGAN #122	
Fluid Density (pcf)	106		Fluid System	CaBr2 Brine	
Hole Size (inch)	Completion Phase				
Casing Depth (m)	3909.5		Solids Control Equip Eff.%		
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)	4208	
Liner Depth (m)	4208		Liner Lap (m)	3782	
Liner OD / ID (inch)	7	6.276	Liner Length (m)	426	

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	250
Cased Hole Volume [Capacity of 3782m 9⅝" Casing + Capacity of 426m 7" Liner]	960
Total New Volume Required (bbl)	**1210**

Estimated Chemical Consumption and Cost based on 74-75pcf MW

PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Calcium Bromide (14.2 ppg)	55 Gal	N/A	0.984bbl/bbl	910	62	972	N/A
Corrosion Inhibitor	55 Gal	N/A	1% Vol	9	1	10	N/A
H2S Scavenger Liquid	55 Gal	N/A	0.5% Vol	5	3	8	N/A
Oxygen Scavenger CA	25 Kg	N/A	0.5 ppb	11	6	17	N/A

*Above calculation are based on normal hole condition

*Above Calculated is based on **106pcf**

7.1.2.9 Estimated contingency material

During all the phase keep a minimum safety stock of below items:

Chemical Type	Unit	Qty	Function
$CaCO_3$ M & C	1 MT BB	10 of each	Loss Circulation Material
Pipe-Freeing Agent	55 Gal DM	4	Environment-Friendly Stuck Pipe Agent
Lime	25 Kg Sx	80	CO_2 Contamination
Zinc Carbonate	25 Kg Sx	120	H_2S Scavenger
Sodium Bicarbonate	25 Kg Sx	40	Cement Contamination
Citric Acid	25 Kg Sx	40	Cement Contamination
Defoamer A	55 Gal DM	4	Anti-Foam
Bio-Cide	55 Gal DM	4	Anti-Bacteria
Oyster shell F/M/C	25 Kg Sxs	160each	Loss circulation material
Drilling Detergent	55Gal	4	Bit Balling Reducer
Oxygen Scavenger S	25Kg	30	Oxygen Scavenger, Polymer Extender
pH Buffer Agent	55Gal	4	Polymer Temperature Stabilizer
Corrosion Inhibitor	55Gal	4	Corrosion Inhibitor
Lubricant	55Gal	8	Biodegradable lubricator

7.1.2.10 Material requirements for each well

Item	QTY	Unit	Description	Function
1	972	55 Gal Dr	Calcium Bromide (14.2 ppg)	CaBr2 brine
2	55	1 MT	Salt	Weighting Agent
3	224	1 MT	Calcium Carbonate 75 Mic	Weighting Agent

4	32	25 Kg Sx	Caustic Soda	PH Control
5	20	50 Kg Sx	Soda Ash	Hardness reducer
6	25	25 Kg Sx	Xanthan-Gum	Viscosifier
9	80	25 Kg Sx	Lime	PH Control
10	160	25 Kg Sx	Shell Fish	LCM
11	160	25 KG Sx	Limestone Chips	LCM
12	10	55 Gal Dr	Corrosion Inhibitor	Completion Fluid Corrosion Inhibitor
13	8	55 Gal Dr	H2S SCAV	Completion Fluid H2S SCAV
14	17	25 Kg Sx	OX SCAV	Completion Fluid OX SCAV
15	1	55 Gal Dr	BIO-CIDE	Anti-bacterial
16	12	55 Gal Dr	PH BUFFER	PH Stabilizer

7.2 Cement Program

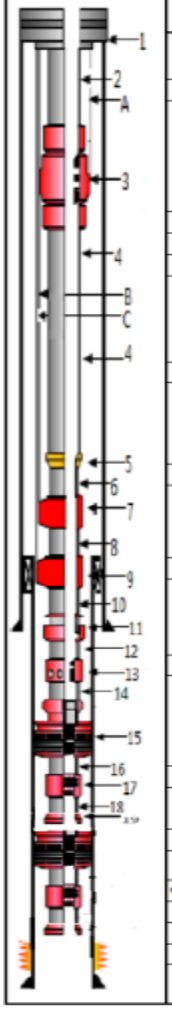
Not applicable.

8. Completion

Completion History:

4-1/2"*3-1/2"-10K Completion string (Weqixin Co.) was run, packer set @ 3930m, try to test completion string several times with no succeed, disconnect completion string from anchor seal & POOH to surface. P/U & M/U 4" anchor seal below first joint & RIH 4-1/2"*3-1/2"-10K completion string (Anton Oil Co.), set packer @ 3832m, perform pressure test was OK, install X-mass tree & test, OK.

8.1 First Run Completion String Which was Failed (Weqixin Co.)

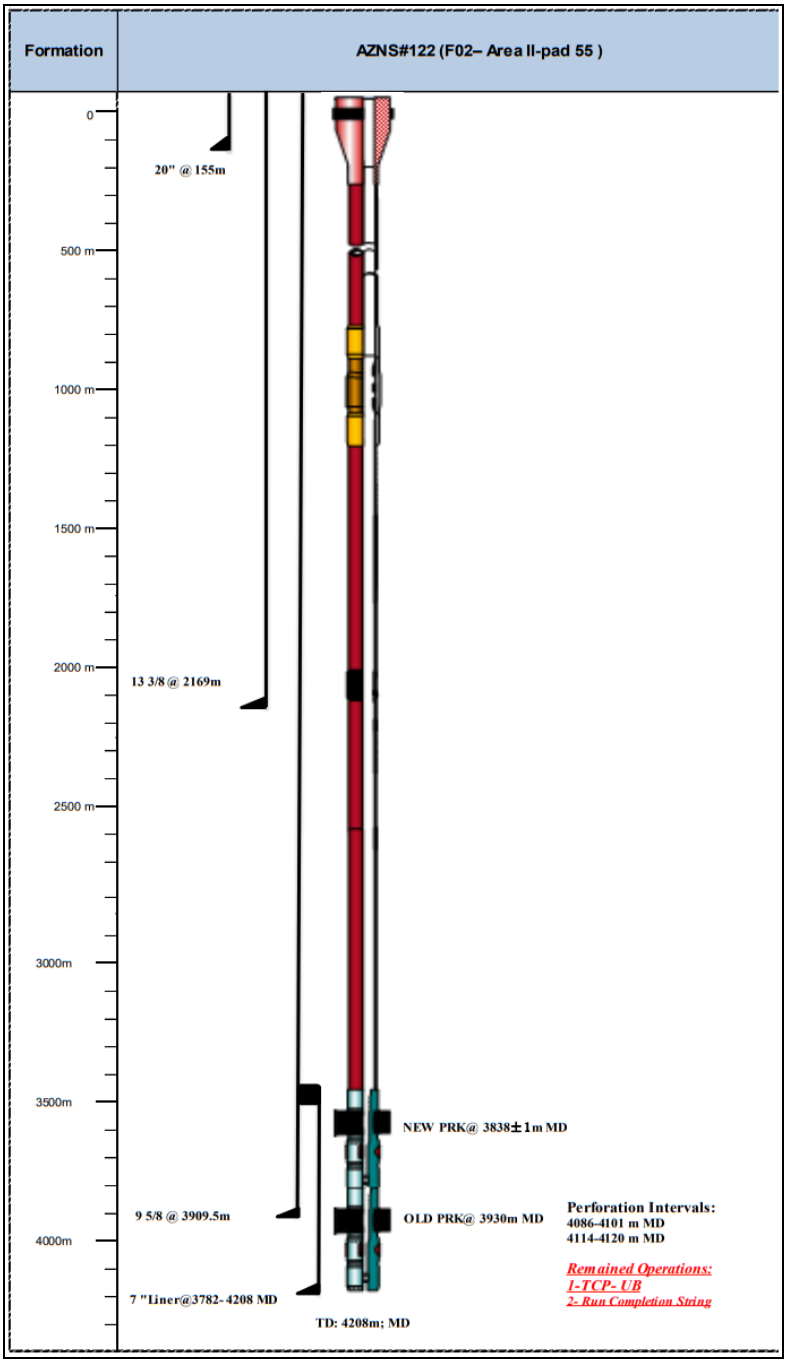


Completion String Schematic of AZNS-122

Item	DESCRIPTION	ID IN	OD IN	LENGTH M	TOP DEPTH M	BOTTOM DEPTH M
1	TUBING HANGER,11" BOWL X 4-1/2"-4 ACME-2G-LH THREAD TOP AND 4-1/2" TPCQ-EX BOX BOTTOM WITH 8-1/2" EXTENDED NECK.	3.640	11.000	0.335	10.765	11.100
	Double-Pin Joint, 4-1/2"18.90 PPF TPCQ-EX P × P	3.640	4.500	1.730	11.100	12.830
2	Tubing, 4-1/2" 18.90 PPF TPCQ-EX BOX x PIN (Qty 9)	3.640	4.500	83.288	12.830	96.118
3	Pup Joint, 4-1/2"18.90 PPF TPCQ-EX B × P	3.640	4.500	1.726	96.118	97.844
	FIOW COUPLING,4-1/2" 18.90 PPF TPCQ-EX BOX	3.640	5.250	1.120	97.844	98.964
	TRSSV-10 TUBING RETRIEVABLE SURFACE CONTROLLED SAFETY VALVE 4-1/2" 18.90 PPF TPCQ-EX BOX x PIN	3.437	7.500	1.737	98.964	100.701
	FIOW COUPLING,4-1/2" 18.90 PPF TPCQ-EX BOX	3.640	5.250	1.120	100.701	101.821
	Pup Joint, 4-1/2"18.90 PPF TPCQ-EX B × P	3.640	4.500	1.746	101.821	103.567
4	Tubing 4-1/2" 18.9PPF TPCQ-EX BOX x PIN(Qty 273)	3.640	4.000	3485.462	103.567	3589.029
5	CROSS-OVER, 4-1/2" 18.90 PPF TPCQ-EX BOX x 3-1/2"10.2ppf TPCQ-EX PIN	2.922	5.250	1.746	3589.029	3590.775
6	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(Qty 29)	2.922	4.000	276.019	3590.775	3866.794
7	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.745	3866.794	3868.539
	1#CHEMICAL INJECTION MANDREL WITH 1.00 'CIV' TUBING RETRIEVABLE CHEMICAL INJECTION VALVE 3-1/2" 10.2# TPCQ-EX Box x Pin	2.922	4.500	0.718	3868.539	3869.257
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.746	3869.257	3871.003
8	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	4.000	9.536	3871.003	3880.539
9	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.745	3880.539	3882.284
	2#CHEMICAL INJECTION MANDREL WITH 1.00 'CIV' TUBING RETRIEVABLE CHEMICAL INJECTION VALVE 3-1/2" 10.2# TPCQ-EX Box x Pin	2.922	4.500	0.718	3882.284	3883.002
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.746	3883.002	3884.748
10	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	4.000	9.536	3884.748	3894.284
11	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.745	3894.284	3896.029
	3-1/2" X 10FT SPLINED EXPANSION JOINT 10.20 PPF TPCQ-EX BOX x PIN	2.922	5.000	5.812	3896.029	3901.841
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.746	3901.841	3903.587
12	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	4.000	9.566	3903.587	3913.153
13	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.745	3913.153	3914.898
	SSD 2.75" NON-ELASTOMERIC SLIDING SIDE DOOR 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN	2.922	5.000	1.195	3914.898	3916.093
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.746	3916.093	3917.839
14	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	4.000	9.561	3917.839	3927.400
15	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.745	3927.400	3929.145
	7" 29-35 PPF CASING HYDRAULIC SET PERMANENT PRODUCTION PACKER WITH SIZE 4.00" ANCHOR SEAL ASSEMBLY 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN	2.922	5.745	1.805	3929.145	3930.950
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.746	3930.950	3932.696
16	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	4.000	9.509	3932.696	3942.205
17	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	4.000	1.745	3942.205	3943.950
	2.562"LANDING NIPPLE 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN	2.635	4.250	0.316	3943.950	3944.266
18	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80.(1 joint)	2.922	4.000	9.516	3944.266	3953.782
19	RE-ENTRY GUIDE SHOE 10.20 PPF TPCQ-EX BOX UP X MULE SHOE GUIDE DOWN	2.900	4.250	0.210	3953.782	3953.992
Other						
A	1/4" X 0.065" X 120 METERS CONTROL LINE					
B	3/8" X 0.065" X 4000 METERS PER ROLL SEAMLESS INJECTION LINE					
C	3/8" X 0.065" X 4000 METERS PER ROLL SEAMLESS INJECTION LINE					

Figure 7: Existing Completion Schematic

8.2 Second & Existing Completion String (Anton Oil Co.)



Item	Name	Description	Remark
1	SIZE 4.00" ANCHOR SEAL ASSEMBLY	SIZE 4.00" ANCHOR SEAL ASSEMBLY SUIT WITH PERMANENT PACKER SUITABLE FOR 3-1/2" 10.20 PPF TPCQ-EX TUBING MIN. ID: 2.922" WORK PRESSURE: 10,000 PSI TEMPERATURE: 325 DEG F MATERIAL: SUPER 13CR-110KSI TOP CONNECTION: 3-1/2" 10.20 PPF TPCQ-EX BOX BOTTOM CONNECTION: MULE SHOE GUIDE TYPE: RATCH LATCH	
2	TOP NO-GO LANDING NIPPLE	MODEL AOF TOP NO GO LANDING NIPPLE SIZE: 3-1/2", WEIGHT: 9.2 PPF, THREAD TYPE: 3-1/2", 9.2 PPF, VAM TOP PIN × BOX, SEAL BORE MIN ID: 2.75" WITH LOCK PROFILE AND SUITABLE LOCK MANDREL OD:3.91" ID: 2.75" WORKING PRESSURE: 10,000 PSI MATERIAL: INCOLOY 925, UNS: N09925 MIN YIELD STRENGTH: 110 KSI	
3	3-1/2" × 7" HYDRAULIC SET RETRIEVABLE PRODUCTION PACKER	MODEL PREMIER PRODUCTION PACKER SIZE: 7"OD:5.820" MIN ID:2.870" CAN BE SET IN 7" 32-35PPF LINER WITH INCLINATION MORE THAN 45° WEIGHT: 32-35 PPF, GRADE: L80, TYPE: 1 AND Q125, TPCQ CASING, 3-1/2", 9.2 PPF, VAM TOP. TUBING CONNECTION: PIN × BOX WORKING PRESSURE: 10,000 PSI MATERIAL: INCOLOY 925, UNS: N09925 MIN YIELD STRENGTH: 110 KSI, DESIGN VALIDATION GRADE: V3 QUALITY CONTROL GRADE: Q1 ELEMENT MATERIAL: AFLAS O-RING MATERIAL: VITON	
4	SSD	MODEL CMD NON-ELASTOMERIC SLIDING SLEEVE FOR SIZE: 3-1/2", WEIGHT: 9.2 PPF, SEAL BORE ID:2.813" THREAD TYPE: VAM TOP CONNECTION: BOX × PIN WORKING PRESSURE: 10,000 PSI MATERIAL: INCONEL 718, UNS: N07718 MIN YIELD STRENGTH: 120 KSI, NON-ELASTOMERIC TYPE. • TO BE RUN INSIDE LINER 7", 32-35 PPF, L80, TPCQ • WITH LOCK PROFILE	

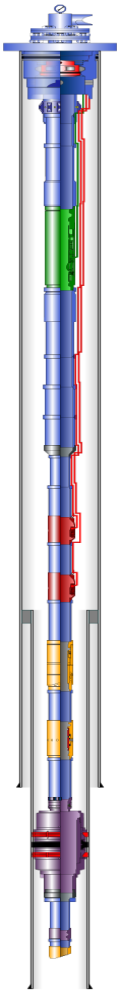
5	KEYED TRAVEL JOINT	MODEL N SPLINED EXPANSION JOINT SIZE: 3-1/2" OD:4.875" ID:2.875" WEIGHT: 9.2 PPF, THREAD TYPE: VAM TOP. CONNECTION: BOX × PIN TOTAL STROKE: 20 FEET, WORKING PRESSURE: 10,000 PSI MATERIAL: INCOLOY 925, UNS: N09925 MIN YIELD STRENGTH: 110 KSI, SEAL MATERIAL: VTR, O-RING MATERIAL: TEFLON/PEEK BACKUP METAL TO METAL SEAL SERVICE: H2S/CO2	
6	CHEMICAL INJECTION MANDREL AND VALVE	ECC CHEMICAL INJECTION SUB OD:5.255" ID:2.949" CAN BE RUN IN 7" 32-35PPF LINER SIZE: 3-1/2",WEIGHT: 9.2 PPF, THREAD TYPE: VAM TOP BOX × PIN WORKING PRESSURE: 10,000 PSI, CONNECTION SIZE: 3/8" MATERIAL: INCOLOY 925, UNS: N09925 MIN YIELD STRENGTH: 110 KSI, CONNECTION TO THE INJECTION LINE SHALL BE PRIMARY AND BACK UP METAL TO METAL SEAL. • ISO 14310 V3 BARRIER SHALL BE QUALIFIED. **CHEMICAL INJECTION VALVE:** MATERIAL: INCOLOY 925 RUPTURE DISC BURST PRESSURE: 7,000 PSI±10% PRESSURE RATING: 10,000 PSI	
7	CROSS COUPLING PROTECTOR	SCREW TYPE SUIT FOR 3-1/2", 9.2 PPF, TUBING, SIZE: INCLUDING 3 SLOTS, ONE FOR 1/4" OD CL, TWO FOR 3/8" OD CHEMICAL INJECTION LINE	
8	CROSSOVERS	3-1/2", 9.2 PPF, VAM TOP PIN × 4-1/2", 15.1 PPF, TPCQ-EX BOX, LENGTH: 1.5 M. MATERIAL: INCOLOY 925, UNS: N09925 MIN YIELD STRENGTH: 110 KSI,	
9	CROSS COUPLING PROTECTOR	SCREW TYPE SUIT FOR 4-1/2", 15.1 PPF, TPCQ–EX TUBING, SIZE: INCLUDING 3 SLOTS, ONE FOR 1/4" OD CL, TWO FOR 3/8" OD CHEMICAL INJECTION LINE	
10-1	SHORT FLOW COUPLING	SIZE: 4-1/2" WEIGHT: 15.1 PPF, LENGTH: 6 FT. THREAD TYPE: 4-1/2", 15.1 PPF, VAM TOP BOX × 4-1/2", 15.1 PPF, TPCQ-EX PIN, FLOW COUPLING ID: 3.826", FLOW COUPLING OD: 5.350" MATERIAL: INCOLOY 925, UNS: N09925	

10-2	TUBING RETRIEVABLE SUBSURFACE SAFETY VALVE	MODEL THUNDER-10E SIZE: 4-1/2 MIN BORE: 3.688", MAX OD: 7.403", WORKING PRESSURE: 10,000 PSI W T: 20-300 °F SETING DEPTH:1000FT OPENING PRESSURE:2000PSI+SHUT IN PRESSURE CONNECTION: VAM TOP MATERIAL: INCONEL 718, UNS: N07718 MIN YIELD STRENGTH: 120 KSI, FEATURES: SELF EQUALIZING AND FLAPPERTYPE, FLAPPER MATERIAL:INC 718 SPRING MATERIAL:X-750 CLASS OF SERVICE:3S	
10-3	LONG FLOW COUPLING	SIZE: 4-1/2" WEIGHT: 15.1 PPF THREAD TYPE: 4-1/2", 15.1 PPF, VAM TOP PIN ×4-1/2", 15.1 PPF, TPCQ-EX, BOX LENGTH: 20 FT FLOW COUPLING ID: 3.826", FLOW COUPLING OD: 5.350". MATERIAL: INCOLOY 925, UNS: N09925 MIN YIELD STRENGTH: 110 KSI,	
10-4	PROTECTOR FOR TR-SCSSSV	(SCREW TYPE) SUIT FOR 4-1/2" TRSCSSSV, SIZE: INCLUDING 3 SLOTS, ONE FOR 1/4" OD CONTROL LINE, TWO FOR 3/8" CHEMICAL INJECTION LINE,	
11.1	4-1/2" NIPPLE	4-1/2", 15.1 PPF, TPCQ-EX, PIN × PIN, LENGTH: 1.5 M MATERIAL: INCOLOY 925, UNS: N09925 MIN YIELD STRENGTH: 110 KSI,	
11.2	4-1/2" NIPPLE	4-1/2", 15.1 PPF, VAM TOP, PIN × TPCQ-EX PIN, LENGTH: 1.5 M MATERIAL: INCOLOY 925, UNS: N09925 MIN YIELD STRENGTH: 110 KSI,	

8.3 Proposed Completion string

AZNS-122 Completion Sketch

Contractor		Liner & Tubing Data					Client	
Field	AZADEGAN	Description	7"Liner	3 1/2" Tubing	4 1/2" Tubing	Tubing	7" Liner Lap:	3782
Well Number:	AZNS-122	OD[in]	7.00	3.50	4.50		7" Liner Shoe:	4208
Rig :	208	Nominal weight [lb/ft]	29.00	10.20	18.90		5' Liner Lap:	-
RTE-GLE	14/45m	Grade	110SS	L-80	L-80		5" Liner Shoe	-
RTE-THF	11/33m	Thread	TPCQ	HSM3	HSM3		Total Depth:	4208
Rig Spudded Date :		Inside diameter [in]	6.184	2.92	3.640		X-mass Tree:	MSP-T TYPE
End of Operation		API Drift Diameter	6.059	2.80	3.515		Size & Rating:	4 1/16"-10K

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)				11.69	0.000
	2	TBG HANGER 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type "H" Two Way	11	3.68	0.6	12.29	11.690
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI	6.1	3.64	1.01	13.300	12.290
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.64	60.940	13.300
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	62.560	60.940
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT	5.5	3.64	1.72	64.280	62.560
	7	TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type,	7.5	3.437	1.78	66.060	64.280
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT	5.5	3.64	1.11	67.170	66.060
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	68.790	67.170
	10	379 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	3605.427	3674.217	68.790
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1.02	3675.237	3674.217
	12	7 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	66.5	3741.737	3675.237
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3743.557	3741.737
	14	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3744.287	3743.557
	15	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3746.107	3744.287
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3755.607	3746.107
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3757.427	3755.607
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3758.157	3757.427
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.8	3759.957	3758.157
	20	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3769.457	3759.957
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3771.277	3769.457
	22	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS,	5.135	2.922	5.755	3777.032	3771.277
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3778.854	3777.032
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3788.354	3778.854
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.625	3789.979	3788.354
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4.551	2.75	1.19	3791.169	3789.979
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.7	3792.869	3791.169
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.27	3802.139	3792.869
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.725	3803.864	3802.139
	30	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD	4/00	2.922	0.23	3804.094	3803.864
	31	7" 29 PPF CASING HYDRAULIC SET RERMANENT PRODUCTION PACKER TO BE RUN IN 7.00" 29 PPF CASING	5//910	2.922	4.44	3808.534	3804.094
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.250	2.922	1.822	3810.356	3808.534
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3819.856	3810.356
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.825	3821.681	3819.856
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.562" TYPE XN (NO-GO ID: 2.442")	4.5	2.562	0.37	3822.051	3821.681
	37	3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI	4.25	2.922	0.24	3822.291	3822.051
		END OF COMPLETION STRING					3822.291

Tubing Retrievable Saftey Valve	66.060	64.280
Traveling Joint	3777.032	3771.277
CIV 1	3744.287	3743.557
CIV 2	3758.157	3757.427
SSD	3791.169	3789.979
7" HYDRAULIC SET PERMANENT PRODUCTION PACKER	3808.534	3804.094
NO-GO Landing Nipple	3822.051	3821.681
Entry Guide	3822.291	3822.051

Figure 7: Proposed Completion Schematic

9. BOP, Wellhead & X-Mass Tree

9.1 Existing Wellhead & X-Mass Tree



Figure 8: Existing Wellhead

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Sarvak	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Sarvak & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Table 10: AZNS Wellhead Information

- **Wellhead Specification**

NOTE: Wellhead and X-Mass Tree already installed by client.

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm**

	Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Table 11: AZNS 122 Wellhead Specification

9.2 BOP

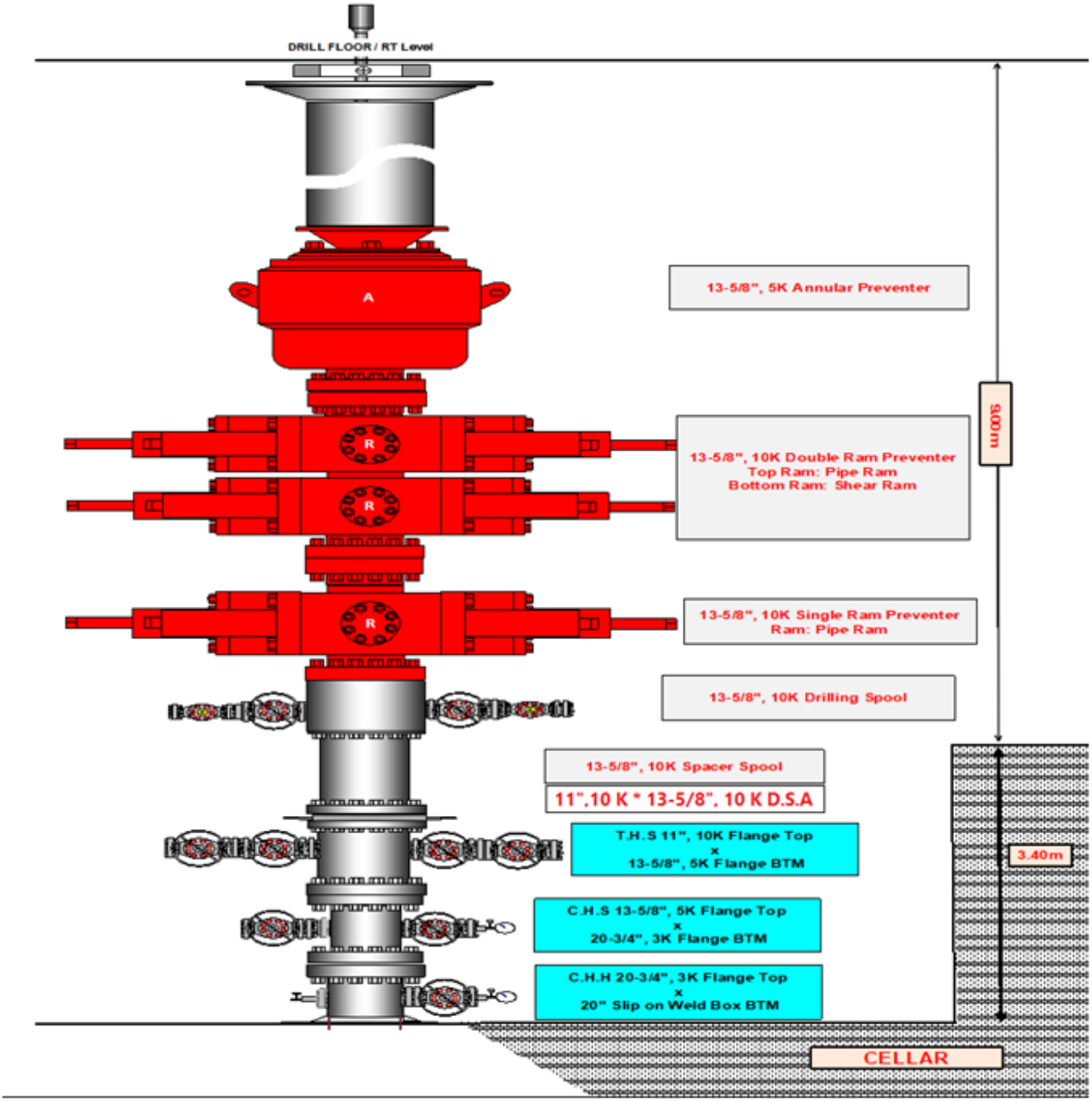


Figure 9: BOP Sketch

10.

10. Waste Management

There is no drilling objective and drilling cutting in AZNS-122 workover well; therefore waste management program is not applicable.

In case of requirement to dispose wastewater in workover well, waste management program will be issued at due time as PEDEC policy.

11. Health, Safety and Environments

12.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA and contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

12.1.1 WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

12.1.2 Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

12.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System and well control equipment shall be recorded in the IADC daily reports and shall be signed by both the Drilling Contractor's Tool Pusher and Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

12.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.
While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

12.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.
Extreme care shall be taken to monitor mud volume, drilling breaks and gas cut mud.
The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.
The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.
The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.
The Driller is responsible for carefully measuring and recording the slow pump pressure.
Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure and record the slow pump pressure at three different flow rates with all pumps at the following times:

Once per tour.

Whenever changes occur in the mud density/rheology.

Whenever changes occur in the dimension and characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

12.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop and a static monitoring of the well shall be carried out. Space-out for all Pipe Rams and Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer and rams.

12.6 Well control during drilling and completion operation

The drilling design shall be strictly implemented by the drilling contractor and when it is discovered there is incompliance of downhole situation with design and the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, and safe operation can be affected, after the discussion between drilling rig manager and the drilling supervisor, practical technical measures can be taken first and then report to the top management. The Company shall approve and notice in time as per the drilling program amendment procedure.

The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program and curve of actual mud weight must be drawn during drilling operation. For exploratory well and key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone and shale shall be drawn by the comprehensive mud logging crew.

When entering oil and gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing and mud loss.

SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate and circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate and circulating pressure shall be recorded in Daily Drilling Report and Kill Sheet.

Once for every 12 hours in general;

After bit change or BHA change; 3) Mud weight change exceeds $^{3.75}$pcf.

Oil and gas invasion and overflow shall be checked with wiper trip and shall be conducted under the following situation:

Each time before POOH after drilling oil and gas formations.

Severe invasion of oil and gas occurs during drilling, but overflow does not occur before POOH.

POOH after killing well.

POOH after lowering mud weight in wellbore.

POOH after curing mud loss during drilling oil and gas formation.

POOH after drilling oil and gas formation and a prolonged period of time is needed for other operations (E logging, running casing, running tubing and DST etc.).

Regarding the shut-in methods selection

Soft shut-in method is recommended for gas well;

Hard shut-in method is recommended for oil well;

Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;

Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.

Flow check shall be conducted regularly via trip tank as follows

Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)

POOH to last casing shoe

Before POOH BHA through BOP stack

Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.

During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.

When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.

During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.

During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. if equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.

Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud. Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.

The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.

During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.

"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if

special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m^3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

12.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.

Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

12.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.

Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.

All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)

All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.

PSSR (Pre-Startup Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

12.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.

Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.

Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.

Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.

Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)

Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.

Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.

The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.

Local community are recommended.

Hazardous Wastes

Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site

Mud purifying circulation system.

Cooling water spraying and circulating system for drilling pump.

Special recovery tank for deposed oil.

Reserve house for valuable mud chemicals.

Automatic mud weighting supply system.

BOP stack is equipped according to drilling design.

Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location

Sewage pit Area of the sewage pit: 1500 m2.

Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.

Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.

Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.

Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.

Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.

After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.

Install fuel supply lines and keep them in good condition.

All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.

The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:

Paved, fenced, marked and illuminated

Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).

In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).

Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).

Capable enough to provide storage for the entire range of hazardous wastes generated during the work.

The area shall be provided with Spill Containment and means of preventing run-off to and from the area.

All waste containers to be stored shall be appropriately labelled.

Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

12. H_2S Monitoring & Prevention Measure

13.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

13.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

13. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.

14. Requirements

Item	Description	QTY.	Unit	Supplier
1	5-7/8" Milled Tooth Bit	1	ea.	MSA
2	5-7/8" Mill	1	Ea.	MSA
3	Taper Tap Suitable for MODEL PREMIER PRODUCTION PACKER SIZE: 7"OD:5.820" MIN ID:2.870"	1	Ea.	MSA
4	Spear Suitable for MODEL PREMIER PRODUCTION PACKER SIZE: 7"OD:5.820" MIN ID:2.870"	1	Ea.	MSA
5	4 1/2" * 3 1/2" Completion String 10K	1	set	MSA
6	9-5/8" & 7" Scraper	1+1	ea.	OEOC
7	5" DP,HWDP,8-1/2" DC,	as required		OEOC
8	Well Head Services	as required		Service Co.
9	Surface Well Test Services	as required		Service Co.
10	Completion Services	as required		Service Co.

11	Slick line Services	as required		Service Co.
12	Casing Running Services	as required		Service Co.
13	Wireline Logging	as required		Service Co.
14	Dry Test, DST, RTTS	as required		Service Co.
15	Cementing Services	as required		Service Co.
16	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
17	H2S Services	as required		Service Co.
18	Mud Logging Services	as required		Service Co.
19	Rig Services	as required		Service Co.
20	Drilling & Completion Fluid Services	as required		Service Co.

Note: Quantity & Spec to be conformed with contract's SOW.
Note: In case of any shortage or discrepancy, the contract's SOW will be reference.

Table 12: List of Requirements

15. Revision Corrections

		• با توجه به عدم امکان ایزولاسیون مطمئن راهکار های مورد نظر این شرکت جهت دریافت رهنمون طی نامه رسمی اعلام گردید.	CM	صفحه ۲۵: بخش Sequence of Operation بند ۱۲ و جدول TBD : - از دیدگاه این مشاور انجام Positive Test نمی تواند اهداف کارفرما انجام Dry Test از لبه آستری ۷اینچ را محقق نماید. لذا مورد تایید باشد.
		• پیرو نامه شماره ۱۴۷۹۲۴ مورخ ۱۴۰۲/۱۱/۰۴ درخواست جلسه در خصوص حل موضوع مذکور گردید.	CM	صفحه ۲۵و صفحه ۱۹ Well Schematic و TBD : - حفاری و بازیابی پکر پایینی به منظور دستیابی به عمق نهایی چاه می بایست انجام گردد.. اصلاحات لازم در Sequence of Operation و TBD برنامه انجام گردد.
		• اصلاح گردید.	CM	Well Montage صفحه ۱۶ واضح و خوانا نمی باشد.
				صفحه ۴۴: عمق نصب پکر پس از لاگ کروژن نهایی خواهد شد.
		• اصلاح گردید.	CM	سرسازندهای شکل صفحه ۱۸ و ۱۹ با شرح کار اختلاف دارد.
		• جدول شماره ۴ مطابق با اطلاعات **Comment/Reply** **Sheet** 00 تهیه گردیده است. اطلاعات موردنظر تحت عنوان جدول شماره ۵ مطابق شرح کار نهایی به برنامه	CM	اطلاعات فشار و دمای مخزنی در جدول ۴ صفحه ۱۳ مطابق با شرح کار ارائه شود (فش دمای تمام زونهای فهلیان مطابق با شرح کار ذکر گردد)

		الحاق گردید.		
		• اصلاح گردید.	CM	برنامه Logging Program در یخش Remark، Before/After Perforation حذف گردد.